Artificial intelligence-based approaches to augmenting and automating surgical training.

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Abstract

To review recent advances in the use of artificial intelligence (AI) to address shortcomings in assessing and improving surgical performance/training by automating surgical skills assessment and feedback. We searched PubMed for studies published between 2015 and 2025 pertaining to AI for surgical training. Search terms included 'artificial intelligence or 'machine learning' or 'deep learning' and 'surgical feedback' or 'surgical training' or 'surgical skill'. Articles were identified with special attention given to those published in the last 5 years with a focus on AI for surgical skill assessment or feedback. Artificial intelligence has been used to successfully automate surgical skill assessment across a variety of surgical disciplines via approaches such as kinematics, sabermetrics, computer vision, and gesture analysis. Many of these studies have developed AI models capable of a binary classification of skill (novice vs expert), which demonstrate concordance when verified against ground truths from human raters. Based on these skills assessments, AI approaches may be further leveraged to generate automatic feedback, which has proven effective in improving surgeon performance metrics, particularly for underperformers. AI has also shown utility in categorising and analysing the content and impact of live surgical feedback, enabling more efficient analysis of how feedback can be best delivered to trainees. Artificial intelligence is a promising tool for augmenting surgical training and improving the objectivity and scalability of surgical skill assessment and feedback. To date, AI models are adept at detecting relatively large differences in surgical performance and providing rudimentary feedback. Further work is required to create models capable of doing more fine-tuned skill assessments and generating more detailed, constructive feedback.

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